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[21]: # Import required libraries
import pandas as pd
from sklearn.linear_model import LogisticRegression

# -----
# Step 1: Create the dataset
# -----
data = {
    'Study_Hours': [1, 2, 3, 4, 5],
    'Result': [0, 0, 0, 1, 1]
}

df = pd.DataFrame(data)

# Display dataset
print("Dataset:")
print(df)

# -----
# Step 2: Separate Features (X) and Target (y)
# -----
X = df[['Study_Hours']]      # Independent variable
y = df['Result']             # Dependent variable
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# -----  
# Step 3: Create Logistic Regression Model  
# -----  
model = LogisticRegression()  
  
# -----  
# Step 4: Train the Model  
# -----  
model.fit(X, y)  
  
# -----  
# Step 5: Display Learned Parameters  
# -----  
print("\nLearned Parameters")  
print("-----")  
print("Weight (Coefficient) =", model.coef_[0][0])  
print("Bias (Intercept)      =", model.intercept_[0])  
  
# -----  
# Step 6: Predict Training Data  
# -----  
predictions = model.predict(X)  
probabilities = model.predict_proba(X)
```

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# -----
# Step 7: Display Results
# -----
print("\nPrediction Results")
print("-----")

for i in range(len(df)):
    print(f"Study Hours = {X.iloc[i,0]} | "
          f"Actual = {y.iloc[i]} | "
          f"Predicted = {predictions[i]} | "
          f"Probability of Pass = {probabilities[i][1]:.4f}")

# -----
# Step 8: Predict for a New Student
# -----
new_student = [[3.5]]

prediction = model.predict(new_student)
probability = model.predict_proba(new_student)

print("\nNew Student Prediction")
print("-----")
print("Study Hours :", new_student[0][0])
print("Predicted Class :", prediction[0])
print("Probability of Pass :", round(probability[0][1],4))

```